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Unit – 5 : Session – 10 : SLO – 2 | LDM/STM and RISC vs CISC

■ I. Multiple Choice

1. C) STM — Store Multiple saves a block of registers to memory.
2. B) RISC — RISC architectures typically use fixed-length instructions.
3. B) Write-back — '!' updates the base register after the transfer.

■ II. Fill in the Blanks

1. Load — LDM = Load Multiple registers.
2. variable-length — CISC uses variable instruction size.
3. stack — STMFD (Store Multiple Full Descending) is the standard stack-push operation.

■ III. True or False

1. TRUE.
2. FALSE — CISC supports more addressing modes than RISC.
3. TRUE — write-back (!) updates the base register automatically.

■ IV. Match the Following

- A) RISC — 3) Simple fixed-length instructions.
B) CISC — 2) Complex instructions.
C) LDM — 1) Loads multiple registers.
D) STM — 4) Stores multiple registers.

■ V. Numerical Answers

1. 12 bytes — 3 registers × 4 bytes each.
2. 24 bits — $32 - 8 = 24$ bits left for operands.
3. STMFD SP!, {R1–R3} with SP = 0x1000: a full-descending push of 3 words decreases SP by 12 → new SP = 0xFF4 (data stored at 0xFF4–0xFFF).